

Yaoyu He

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Education

ShanghaiTech University, Shanghai, China , Computer Science	2022.09 - 2026.07
- major B.E. in Computer Science Coursework: Computer Architecture(GPA: 4.0), Machine Learning(GPA: 4.0), Deep Learning(GPA: 4.0), Artificial Intelligence(GPA: 4.0), Computer Vision(GPA: 4.0)	
ShanghaiTech University, Shanghai, China , Computer Science	2026.09 - 2031.07 (expected)
major Ph.D. in Computer Science	

Research Interests

My research focuses on Embodiment AI and Multimodal LLM.

- Embodiment AI: Reinforcement Learning, Imitation Learning, Robotics, Multimodal Perception
- Multimodal LLM: Large Language Model, Multimodal Reasoning, Prompt Engineering

Publications

Sophia-in-Audition: Virtual Production with a Robot Performer	2024.02
Taotao Zhou, Teng Xu, Dong Zhang, Yuyang Jiao, Peijun Xu, Yaoyu He , Lan Xu, Jingyi Yu <i>Proceeding of the 32nd ACM International Conference on Multimedia</i> 2402.06978 🔗	
Mojito: LLM-Aided Motion Instructor with Jitter-Reduced Inertial Tokens	2025.02
Ziwei Shan, Yaoyu He , Chengfeng Zhao, Jiashen Du, Jingyan Zhang, Qixuan Zhang, Jingyi Yu, Lan Xu <i>arxiv preprint</i> 2502.16175 🔗	
From Noise to Intent: Anchoring Generative VLA Policies with Residual Bridges	2026.01
Yiming Zhong, Yaoyu He , Zemin Yang, Pengfei Tian, Yifan Huang, Qingqiu Huang, Xinge Zhu, Yuexin Ma <i>Forty-Third International Conference on Machine Learning</i> 2604.21391 🔗	

Projects

From Noise to Intent: Anchoring Generative VLA Policies with Residual Bridges	ResVLA 🔗
- Proposed ResVLA, a novel generative visuomotor policy framework that shifts from "generation-from-noise" to "refinement-from-intent", addressing inefficiencies in existing VLA policies. - Introduced a spectral decomposition strategy to decouple low-frequency semantic intent and high-frequency action dynamics, enabling structured action generation via residual diffusion bridges. - Achieved strong performance across diverse benchmarks, demonstrating improved robustness in long-horizon tasks and enhanced cross-embodiment generalization.	
Mojito: LLM-Aided Motion Instructor with Jitter-Reduced Inertial Tokens	Mojito 🔗
- Using VQ-VAE-based tokenizer, mitigating noise and drift - Using Transformer-based generator, generating advisor based on the input of inertial tokens and text instructions	
Chinese Checkers AI Agent	Chinese Checkers 🔗
Developed hybrid RL agent combining Approximate Q-learning with Monte Carlo Tree Search	

- Implemented neural heuristic evaluator using self-play reinforcement learning

Coursebench

[coursebench-frontend](#) 

- Develop a website about the course selection page

Research and Work Experience

LumiAni Technology Co., Ltd Developer and Researcher

2023.06 - 2024.07

- Developed and optimized stable diffusion plugin for 3D reconstruction pipeline
- Apply Colmap for camera calibration and 3D Gaussian Splatting for object point cloud reconstruction, and compared with NeRF/NeuS2
- Design LLM-based prompt engineering strategies to automate labeling and analysis of rodent behavior data

Shanghai Elan Smart Sense Information Technology Co.,LTD.

2024.07 - 2025.01

- Independently responsible for the construction and debugging of LLM modules, using Qwen, flan-t5, gpt2 and other large language models for debugging. Implemented a conversion module that converts IMU semantic information into LLM semantic space.
- Use LoRA to fine-tune and optimize the large language model, and use DPO human preferences to optimize the output for the large language model
- Processing 30,000+ data entries by aligning text, SMPL motion, and IMU data while performing IMU data cleaning
- Lead the website construction part, independently design the website architecture, and lead other students to complete the website construction

Shanghai Jiaotong RL² lab

2025.02 - 2025.08

- Integrating the G1 humanoid robot into the Robosuite framework with a three-fingered gripper implementation
- Implement PPO with curiosity-driven exploration for robot skill acquisition, enabling autonomous adaptation to unseen physical environments through intrinsic reward mechanisms aligned with embodied intelligence principles
- Investigate VLA for semantic-aware robotic grasping, leveraging multimodal reasoning to interpret open-vocabulary commands and improve task generalization in real-world scenarios

ShanghaiTech 4DV Lab

2025.08 - Now

- Work on reinforcement learning with piper robot arm, aim to rotation basketball
- Deploy and fine-tuning $pi_{0.5}$ model on libero robot

Technologies

Languages: Python, C++, C, JavaScript, Typescript, Rust, Java, C#,

Technologies: PyTorch, Mujoco, Gym, Qt, Unity, React, Vue, SpringBoot, Tauri